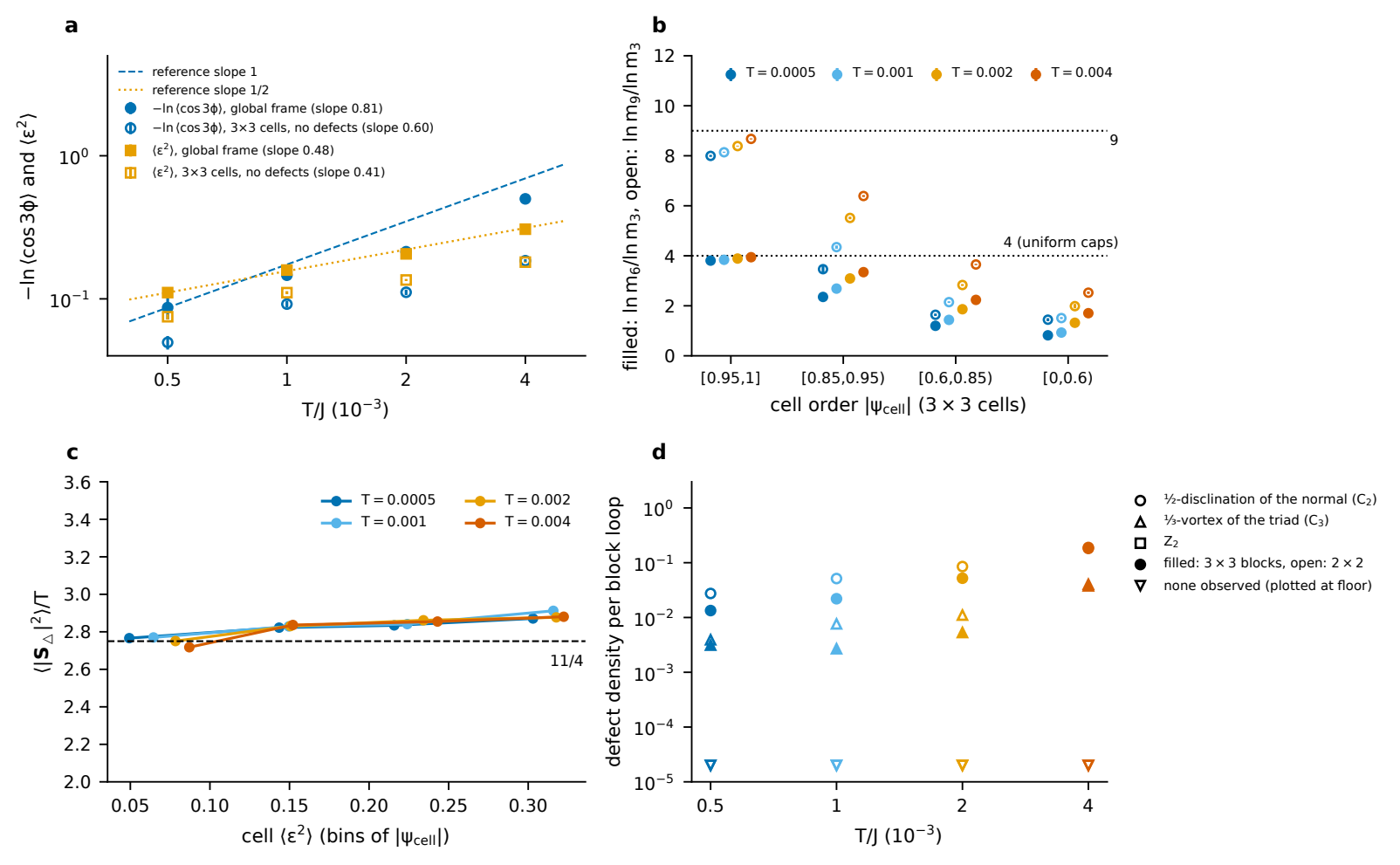
a
reference slope 1
reference slope 1/2
−ln ⟨cos 3ϕ⟩, global frame (slope 0.81)
−ln ⟨cos 3ϕ⟩, 3×3 cells, no defects (slope 0.60)
⟨ε²⟩, global frame (slope 0.48)
⟨ε²⟩, 3×3 cells, no defects (slope 0.41)
−ln ⟨cos 3ϕ⟩ and ⟨ε²⟩
T/J (10⁻³)
b
T = 0.0005
T = 0.001
T = 0.002
T = 0.004
9
4 (uniform caps)
filled: ln m₆/ln m₃, open: ln m₉/ln m₃
[0.95,1]
[0.85,0.95)
[0.6,0.85)
[0,0.6)
cell order |ψcell| (3 × 3 cells)
c
T = 0.0005
T = 0.001
T = 0.002
T = 0.004
11/4
⟨|S△|²⟩/T
cell ⟨ε²⟩ (bins of |ψcell|)
d
defect density per block loop
T/J (10⁻³)
½-disclination of the normal (C₂)
⅓-vortex of the triad (C₃)
Z₂
filled: 3 × 3 blocks, open: 2 × 2
none observed (plotted at floor)